KREAYOTOO SOLUTIONS (P) LTD

Amaravati Capital City

Underground Utility Digital Twin

Project Report — What We Built, How It Works & Where We Are Going

Andhra Pradesh Capital Region Development Authority (APCRDA) · Amaravati, Andhra Pradesh

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78,053 pipes	43,685 chambers	95,935 plots	21 blocks	6 zones	5 utilities

1. PROJECT OVERVIEW

Kreayotoo Solutions (P) Ltd (KSPL) has built a **Digital Twin of Amaravati Capital City's underground utility infrastructure** for the Andhra Pradesh Capital Region Development Authority (APCRDA). The twin covers the complete underground network — water supply, sewerage, storm-water drainage, reuse water and power — across 21 blocks in 6 zones of the planned capital city.

The deliverable is a **single offline HTML application** (double-click to run, no server needed) that provides 3D visualisation, GIS mapping, BIM building integration, live weather and storm simulation, and an AI-assisted decision support system driven by a virtual SCADA sensor network. The platform is designed to evolve from a visualisation tool into a full operational system as real IoT sensors and SCADA are connected.

Project Objectives

- Provide a unified 3D+GIS view of all underground utilities across Amaravati's 21 blocks
- Enable zone-by-zone and city-wide network analysis
- Integrate real BIM building data (Revit IFC) with the infrastructure
- Simulate storm-water network response to live rainfall
- Build an alert-driven decision support system with real GIS buffer analysis
- Establish a foundation ready for live SCADA/IoT sensor integration
- Demonstrate the SIWA-style operational capability (Siemens benchmark)
- Ensure honest labelling: real data vs simulated always distinguished
- Create a reusable platform engine for other cities and airports
- Deliver a fully self-contained offline package (no internet needed for core)

Project Scope — What Is Covered

Area	Amaravati Capital City — 21 blocks across 6 zones (Zone 1B, 5A, 5C, 5D, 10, 12)
Utilities	Water supply · Sewerage · Storm-water drainage · Reuse water · Power/electrical conduits
Network scale	78,053 pipes · 43,685 manholes/chambers · 2,131+ km of utility mains · 117 km trunk main
BIM	WTP Admin Building — 16-storey Revit IFC4 (architectural + structural), real materials, 89 rooms
Masterplan	95,935 plots from the APCRDA masterplan shapefile with full landuse zoning (35 zone types)
Weather	Live rainfall (Open-Meteo) + Windy.com radar, driving simulated storm-water network response
Decision Support	Alert-driven with real GIS buffer (95,935 plots + 78,053 pipes), 5 scenarios, virtual SCADA sensors

2. WHAT WE BUILT — MODULES

3D Network Twin

● BUILT

Dark SCADA-style 3D viewer — individual block / zone / entire city views. Free pan/orbit/zoom. Click any pipe or manhole to see its attributes.

GIS Map

● BUILT

Interactive Leaflet map with satellite basemap, all 5 utility layers coloured by type, trunk main polylines, block markers, and tap-to-locate.

Landuse Zoning Overlay

● BUILT

All 95,935 masterplan plots rendered as a georeferenced colour image (QML colours) — toggleable on both the overview and per-block maps, with a 35-zone legend.

Trunk Main

● BUILT

Dedicated 3D + GIS view of the 117 km trunk water main network across all zones.

WTP Admin Building — BIM

● BUILT

16-storey Revit IFC4 in 3D — real Revit material colours (24 colours, 0% black), floor-by-floor isolation, 89 clickable room labels with area and level, X-ray/ghost/edges modes.

Building Georeferencing

● BUILT

IfcProjectedCRS + IfcMapConversion injected into the IFC — EPSG:32644 (UTM 44N), placed at the network centre. Any GIS/BIM tool will now read the correct position.

Live Weather + Storm

● BUILT

Open-Meteo live rainfall (current, 24h, 7-day) + Windy.com animated radar map. Rolling-window intensity + antecedent wetness drive a real Manning capacity model per zone. Demand and raw-water outlook also updated.

Decision Support

● BUILT

Alert-driven: 5 scenarios (pipe burst, contamination, power failure, heavy rain, fire). Real GIS buffer on 95,935 plots + 78,053 pipes. Impact tab, Network tab, Actions checklist, Timeline chart, Sensors tab, Audit Log. Generates a full incident report.

Virtual SCADA Sensor Network

● BUILT

~50 virtual sensors (pressure transmitters, DMA flow meters, chlorine analysers, pump monitors) across 21 blocks. Readings evolve every 20 s with real diurnal patterns and noise. Faults spawn, escalate and clear. Every alert is a real threshold breach.

Continuous Alert Engine

● BUILT

Feed re-checked every 20 s. New/escalated alerts trigger toasts, auto-follow re-runs impact analysis. ACK button, audit log, export. Sidebar badge reflects live count.

Zone 10 Invert Fix

● BUILT

6,784 manholes in Zone 10 (blocks B3, B7, B8) had zero invert levels (data gap) — derived from connected pipe inverts. Depths now 1.2–5.8 m (median 2.0 m).

Manning Capacity per Zone

● BUILT

Real storm-drain outfall capacity per zone computed from Civil 3D pipe diameters and invert-derived slopes (Manning's equation, n=0.014).

Isolation Points

● BUILT

Scenario-aware: valve chambers for burst/contamination, power chambers for power failure, storm chambers for flood, hydrant points for fire.

DMA Leak Detection

● REMOVED

Minimum Night Flow + mass-balance panel built and tested — removed at client request, ready to reinstate when needed.

3. HOW IT WORKS — ARCHITECTURE & DATA FLOW

The platform is a **single offline HTML file** (~33 MB) embedding all data, 3D geometry, maps, dashboards and application code. It runs by double-clicking — no server, no installation. Internet is needed only for the live basemap tiles, weather API and Windy radar. The architecture has five layers:

Layer 1 — Source data	Civil 3D pipe networks (xlsx quantities), Revit IFC4 building, APCRDA masterplan shapefile, QGIS QML zoning styles, ECW ortho imagery (pending conversion)
Layer 2 — Build pipeline	Python scripts: parse xlsx → compact JSON per block; tessellate IFC via ifcopenshell → Draco-compressed glTF (building.glb); render shapefile → georeferenced PNG; extract IfcSpace rooms; fix invert levels; compute Manning capacity. Output: one set of data files.
Layer 3 — make_app.py	Assembles the single HTML file: embeds the dashboard template, all 21 block networks, trunk main, building GLB+floors+rooms, landuse PNG, weather dashboard, decision dashboard (with 95k plots + 78k pipes), weather/storm params, zone storm capacity. All tokens replaced at build time.
Layer 4 — Browser runtime	Three.js (3D), Leaflet + Esri Leaflet (GIS), DRACOLoader (compressed building geometry), Open-Meteo fetch (live weather), virtual sensor simulator (20s tick), real-time GIS buffer query (spatial loop over 95,935 plots + 78,053 pipes — runs in <1s in-browser).
Layer 5 — Live feeds (future)	SCADA/IoT endpoint (ALERT_FEED_URL), Windy Point Forecast API (key needed), DEM for flood depth, EPANET hydraulic model, consumer connection database.

Data Flow: Source → App

Civil 3D / xlsx	→	build_block.py	→	**Block network JSON (pipes, manholes, dims)**
Revit IFC4	→	ifc_to_glb.py + build_ar_floors.py + build_st_combine.py	→	**building_real_draco.glb (floors + colours + rooms)**
Masterplan shp	→	render_landuse.py + build_incident_data.py	→	**landuse.png + plots.json + incident_net.json**
make_app.py	→	Assembles all of the above	→	**Amaravati_Twins_App.html**

4. SIMULATION METHODS — HOW EACH NUMBER IS CALCULATED

4.1 Virtual Sensor Network (demo SCADA)

The twin runs approximately 50 virtual sensors across the 21 blocks — pressure transmitters (PT), DMA flow meters (FM), chlorine analysers (CL) and pump monitors (PU). Readings are recalculated every 20 seconds following a real diurnal demand pattern with noise. Faults spawn randomly, ramp up over 2–8 minutes and clear after 8–35 minutes. Every alert in the feed is a threshold breach on a moving sensor reading — exactly how real SCADA alerting works.

Pressure	value = base × (1.08 – 0.10 × diurnal_factor) + noise(±0.06 bar). Fault: value × (1 – 0.62 × magnitude × ramp_progress). Alert threshold: below 2.35 bar.
Flow	value = base × diurnal_factor + noise(±5%). Fault: value × (1 + 1.25 × magnitude × ramp_progress) — inlet flow spikes as a leak develops. Alert threshold: above +45% of base.
Chlorine	value = base + noise(±0.03 mg/L). Fault: value × (1 – 0.80 × magnitude × ramp_progress). Alert threshold: below 0.20 mg/L (WHO minimum).
Pump	value = base × (0.85 + 0.20 × diurnal) + noise(±4 kW). Fault: value × (1 – magnitude × ramp_progress) — pump draws less power as it fails. Alert threshold: below 35% of base load.

4.2 GIS Buffer Analysis (real spatial query)

When you tap the map, the twin runs a real point-in-circle query against all 95,935 plot centroids and 78,053 pipe mid-points stored as integer metre offsets. No external GIS server is involved — the full computation runs in the browser in under one second.

Plot query	for each plot: distance² = (plot.dx – mx)² + (plot.dy – my)²; if < radius² → include
Population	Population = Σ (plot_area_ha × planning_density_of_zone) e.g. R3 zone = 250 persons/ha · R4 = 400 · Parks = 2 · Roads = 0
Demand	Water demand at risk (MLD) = Population × 135 LPCD ÷ 1,000,000
Road length	Road reserve area (from zoning) ÷ 12 m assumed width
Risk score	score = pop/220 + 4×(hospitals×3 + schools×1.5 + govt) + 1.4×(total km in buffer) × 1.25 for contamination/fire · capped at 100

4.3 Storm-water Response (live rainfall → real capacity)

Live rainfall input	Open-Meteo hourly precipitation (past 48 h + next 24 h). Peak intensity = worst of peak-1h and rolling 3-hour average.
Antecedent wetness	C_effective = min(0.95, 0.70 + min(rain_48h / 60, 1) × 0.22) Saturated ground raises the runoff coefficient from 0.70 toward 0.95.
Runoff (rational method)	Q_peak = C_eff × intensity (mm/hr) × catchment_area (m²) ÷ 3,600,000 [m³/s]
Outfall capacity	Q_cap = (1/n) × A × R^(2/3) × S^0.5 [Manning, n=0.014] Computed from **real Civil 3D pipe diameters and invert-derived slopes**. e.g. Zone 12 (1500 mm) = 1,367 m³/s · Zone 10 (200 mm) = 62 m³/s
Utilisation	Utilisation (%) = Q_peak / Q_cap × 100 Alert: ≥50% = medium, ≥70% = watch, >100% = overflow risk

4.4 Rule Engine — Recommended Actions

The decision engine is a transparent deterministic rule set — not a trained AI model. Every card shows its formula and assumption. Key formulae:

Burst discharge	Q = Cd × A × √(2gH) where Cd=0.62, bore fraction=20%, H=30 m (assumed head) e.g. 600 mm main → Q ≈ 270 L/s ≈ 972 m³/hr
Contaminant travel	Speed = 0.9 m/s = 3.2 km/hr · time to buffer edge = radius ÷ 3,200 (min)
Power / storage	Storage time = 8 hrs (CPHEEO standard) · with 25% demand cut → 10.7 hrs
Fire flow	Required: 30–60 L/s at 2 bar residual. A 300 mm main can deliver; <300 mm may not.
Repair time estimate	Isolation 25 min · Excavation + repair: 3–5 hrs (<600 mm), 6–10 hrs (≥600 mm) Recharge + flush: 1 hr → Total: 5–7 hrs or 8–12 hrs

5. TECHNOLOGY STACK

The entire platform is built on open-source and free-tier tools — no proprietary middleware or licences required for the core twin. Live API services are free/key-free.

Category	Technology / Library	Version / Notes	Role
3D Viewer	Three.js	r128, CDN	WebGL 3D rendering for pipe networks and building
3D Viewer	DRACOLoader	Three.js addon	Decompress Draco-compressed building GLB in browser
GIS Map	Leaflet	1.9.4, CDN	Interactive map — layers, markers, popups
GIS Map	Esri Leaflet	3.0.x, CDN	Satellite basemap (Imagery + ImageryLabels)
Weather	Open-Meteo API	Free, no key	Live temperature, rainfall, 7-day forecast for Amaravati
Weather	Windy.com embed	Free embed	Animated radar map — rain/wind/temp/cloud layers
BIM/IFC	ifcopenshell	0.8.5, Python	Parse IFC4, tessellate geometry, extract rooms
BIM/IFC	trimesh	4.x, Python	Build scene, export GLB, apply transforms
BIM/IFC	gltf-pipeline	npm, Node.js	Draco compression of building GLB (872 KB from 14 MB)
GIS process	pyshp (shapefile)	Python	Read masterplan shapefile (95,935 plots)
GIS process	pyproj	Python	CRS transforms: EPSG:32644 ↔ EPSG:4326
GIS process	Pillow (PIL)	Python	Render landuse colour image (4200×3757 px PNG)
Build	Python 3.12	Build-time	All data processing, build pipeline, PDF report
Build	ReportLab	Python	This project report PDF
Compression	Draco (Google)	Via gltf-pipeline	Building GLB: 14.5 MB → 872 KB (16× smaller)
Coord system	EPSG:32644	UTM Zone 44N	Native CRS for Amaravati — all network data
Coord system	EPSG:4326	WGS84	Web map display, building georeferencing
Standards	IFC4 (ISO 16739)	Revit export	Open BIM standard for the Admin building
Standards	IfcMapConversion	IFC4	Georeferencing: local coords → UTM 44N
Standards	Manning's equation	Engineering	Storm-drain outfall capacity from real pipe data
Standards	CPHEEO guidelines	India water	Storage norms, demand planning (135 LPCD)

6. DATA — REAL vs SIMULATED

A fundamental principle of this platform is honest, unambiguous labelling of data quality. The interface uses colour-coded tags: ● **REAL** (green) for data from your actual Civil 3D / GIS / Revit sources, and ■ **SIMULATED** (amber) for modelled or demonstration values that will be replaced by live sensor data as the SCADA system is deployed.

	Data / Parameter	Status	Source / Notes
●	**Pipe geometry, diameters, lengths, connectivity**	REAL	Civil 3D xlsx quantities → build_block.py
●	**Manhole positions, rim levels, invert levels**	REAL	Civil 3D. Zone 10 zero-inverts derived from connected pipes.
●	**Storm drain capacity per zone**	REAL	Manning's equation on real diameters + invert-derived slopes
●	**Masterplan plot positions, zoning, areas**	REAL	APCRDA shapefile — 95,935 plots, 35 zone types
●	**Landuse zoning colours**	REAL	QML colour scheme from QGIS
●	**Building geometry, floors, materials**	REAL	Revit IFC4 tessellated via ifcopenshell
●	**Building room names and areas**	REAL	IfcSpace objects from Revit
●	**Rainfall / temperature / 7-day forecast**	REAL	Open-Meteo live API (free, no key, auto-refreshed)
●	**Windy live radar map**	REAL	Windy.com embed (free, no key)
●	**GIS buffer spatial query**	REAL	In-browser point-in-circle on real plot + pipe data
■	**Sensor readings (pressure, flow, chlorine, pump)**	PARTIAL	Virtual simulator — real diurnal pattern, noise and fault model. Replaced by SCADA.
■	**Population in the buffer**	PARTIAL	Plot area × HMDA planning density — not census data
■	**Burst discharge**	PARTIAL	Orifice equation — assumed 30 m head, 20% bore. Replaced by live pressure.
■	**Contaminant travel speed**	PARTIAL	Assumed 0.9 m/s. Replaced by live flow meter.
■	**Tank storage / drawdown**	PARTIAL	CPHEEO standard 8 hrs. Replaced by live tank level.
■	**Repair times**	PARTIAL	Engineering estimates. Replaced by historical asset maintenance log.
✗	**Flood depth**	PENDING	Needs a DEM (terrain model) + SWMM hydraulic routing
✗	**Pump station locations**	PENDING	Needs the electrical asset register
✗	**Valve register**	PENDING	Needed for true minimum-valve-set isolation

✗	**Consumer connections**	PENDING	Needed for exact population count and NRW

7. DELIVERABLES

Amaravati_Twins_App.html	~33 MB single-file offline application. Double-click to run. Contains all networks, building, landuse, dashboards and code. Internet needed only for map tiles and live weather.
Amaravati_WTP_Admin_Building.html	~1.3 MB standalone BIM viewer — full building with floor isolation, X-ray, edges, ghost, 89 room labels.
WTP-ADMIN-AR_09_georeferenced.ifc	~32 MB georeferenced architectural IFC4 — IfcProjectedCRS + IfcMapConversion (EPSG:32644) injected. Opens in any GIS or BIM tool at the correct Amaravati location.
DECISION_SUPPORT.md	Full documentation of the decision support system: data flow, all formulae, real-vs-modelled table, 16-module status, alert payload schema, and next-data-needed list.
Amaravati_DigitalTwin_FullPackage.zip	~77 MB complete package: • build_pipeline/ — all Python scripts + HTML templates + logo • datasets/ — all data JSON, building source, landuse, decision datasets • webapp/ — hosted version (Vercel-deployable) • docs/ — DECISION_SUPPORT.md + workflow • README.txt — complete guide

Package Contents Detail

- 21 block network HTMLs (individual twin files)
- Zone combined HTMLs (6 zones)
- City and trunk combined views
- manifest.json (block metadata)
- combined.json (zone+city data)
- Trunk_Main_network.json
- plots.json (95,935 plot centroids + zoning)
- incident_net.json (78,053 pipes + 43,685 chambers)
- blocks_ll.json (21 block lat/lon)
- landuse.png + landuse_bounds.json + legend
- building_real_draco.glb (Draco-compressed building)
- floors.json + floor_order.json + spaces.json
- elem_color.json (real Revit material colours)
- recenter.json (model origin offset)
- bldg_loc.json (georeferenced position)
- Landuse_Styles.qml (original QGIS style)
- Source xlsx workbooks (all 21 blocks)
- Trunk_Model.dxf
- build_*.py, make_*.py, render_*.py scripts
- dashboard_template.html + all dashboard HTMLs

8. BENCHMARK — SIEMENS SIWA vs AMARAVATI TWIN

The Siemens SIWA platform (used in Spain at Servicios de Txingudi and Malaga) is an AI-powered water network analytics suite. Kreayotoo reviewed the SIWA case study to benchmark the Amaravati twin and map a realistic roadmap to that capability level.

Capability	SIWA (Spain)	Amaravati Twin (current)	Gap / Next step
Network model	285 km pipes modelled	2,131+ km pipes in Civil 3D data	EPANET hydraulic model to build
Leak detection	Real MNF on live DMA meters AI pre-locates to 45 m	MNF logic built — needs real flow meters	Install DMA inlet flow meters
Sensor network	Flow, pressure, quality, AMI	~50 virtual sensors (demo)	Install PT/FM/CL sensors + SCADA
Customer meters	238,000 AMI meters (Malaga)	None yet	Smart meter rollout (Phase 3)
AI / ML	Trained on 3+ years of data	Rule engine (transparent, no training data)	ML possible after 1+ year of SCADA data
SCADA historian	InfluxDB / time series	Not yet	InfluxDB + data pipeline (Phase 2)
NRW accounting	Per-DMA mass balance, live	Framework built, needs live meters	DMA meters (Phase 1)
3D / GIS twin	Not the SIWA focus	Full 3D + GIS + BIM (built)	AHEAD of SIWA in this dimension
Offline / portable	Server-hosted	Full offline single-file HTML	AHEAD — unique advantage
GIS buffer analysis	Not in SIWA	95,935 plots + 78,053 pipes, real-time	AHEAD — unique to this twin

Key insight: The Amaravati twin is **ahead of SIWA** in 3D/GIS visualisation, offline portability and spatial decision support. SIWA is ahead in live sensor analytics and AI. The gap is instrumentation — once SCADA and DMA meters are connected, the same analytical framework already built in this twin runs on real data with no redesign.

9. FUTURE ROADMAP

The roadmap is structured in six phases. Phases 0–1 require no new software — only data. Phases 2–6 progressively close the gap to full operational AI-level capability.

Phase	Title	Deliverables / milestones
Phase 0	Foundation (DONE)	• All 21 blocks + 6 zones + city + trunk in the twin • WTP Admin Building BIM (16 floors + 89 rooms + real materials) • Landuse zoning overlay (95,935 plots) • Live weather + storm-water simulation • Alert-driven decision support + GIS buffer analysis • Virtual SCADA sensor network (demo) • Georeferenced IFC (EPSG:32644)
Phase 1	Real Data Integration (now → 6 months)	• ECW ortho imagery: convert to GeoTIFF in QGIS → drop into Zone 5D overlay • WTP Admin Building: surveyed Easting/Northing → exact map placement • Valve / hydrant register → true minimum-valve-set isolation • Electrical asset register → pump stations, substations, DG sets • DEM (terrain model) → flood depth + evacuation routing • Consumer connection database → exact population, NRW per zone
Phase 2	DMA Instrumentation (6–18 months)	• Install DMA inlet electromagnetic flow meters (one per block entry) • Install pressure loggers at zone inlets and network extremities • Install water-quality sensors (chlorine, turbidity, pH) at WTP outlet + key points • Connect to SCADA / IoT platform (MQTT or REST) → set ALERT_FEED_URL • Time-series database (InfluxDB or TimescaleDB) + historian • Leak detection dashboard reinstated on real Minimum Night Flow data
Phase 3	Smart Meters + NRW (18–36 months)	• Smart meter (AMI) rollout — per-connection daily consumption • NRW panel: produced vs billed vs metered per DMA, live mass balance • Demand forecasting: temperature + holiday + historical pattern → next-day peak • EPANET hydraulic model: pressure heatmap, valve operation simulation, pipe failure modelling • Automated supply interruption notification (SMS / app) from the decision engine
Phase 4	AI Analytics (after 1+ year of SCADA data)	• Leak pre-location: ML on pressure/flow patterns → narrowed to 45 m (SIWA level) • Predictive maintenance: pump bearing failure prediction from current + vibration • Burst risk model: pipe age + material + pressure history + break history • Water quality prediction: contamination spread direction + concentration + time • Seasonal demand model trained on accumulated consumption data
Phase 5	Conversational AI + Automation (36+ months)	• AI chat assistant: "Why is pressure low in Zone 5A?" → real answer from live data • Automated valve control recommendation with one-click approval • Digital twin time-travel: replay any historical incident with measured data • Risk map: city-wide colour heatmap updated hourly from sensor health • Full SIWA-level operational capability: leak finder + meter analytics + NRW + quality
Phase 6	Platform Reuse (parallel track)	• Separate the engine (3D/GIS/dashboards) from the project config (CRS, utilities, data paths) • Config-driven project scaffolder: new city = one config file + data folder • Airport variant: apron/taxiway/terminal zones, fuel lines, BMS/HVAC feeds, gates • SaaS hosting: multi-project dashboard with project switcher • First reuse project: existing city or airport client onboarded on the same engine

10. WHAT DATA IS NEEDED NEXT

The following data items, in order of impact on the platform's accuracy and capability, are either already in KSPL's possession or can be obtained from APCRDA/project records without new surveying or instrumentation:

Immediate — no new work needed

- WTP Admin Building surveyed Easting/Northing (from site-layout drawing) → exact building position on map
- ECW ortho for Zone 5D → convert to GeoTIFF in QGIS (File → Export → Save As, EPSG:32644, JPEG, ~6000 px) → send the .tif
- Remaining zone ortho files → same conversion
- Any additional block xlsx workbooks not yet in the package

Short term — from existing records

- Valve and hydrant register (location, type, size, status) → true isolation-point optimisation
- Electrical asset register (pump stations, substations, DG set locations, ratings)
- DEM / terrain model for Amaravati (even a 5 m resolution SRTM or LiDAR extract)
- Consumer connection database (plot ID, meter ID, monthly consumption)

Medium term — from instrumentation

- SCADA: DMA inlet flow meters + pressure loggers per block
- Water quality sensors: chlorine residual, turbidity, pH at WTP + key network points
- Tank level sensors + pump telemetry (flow, kW, status)
- Any existing SCADA data extracts (even CSV) from temporary monitoring campaigns

For SCADA integration

- Set ALERT_FEED_URL in decision_dashboard.html to your SCADA REST/MQTT endpoint
- Payload: id, type, severity, lat, lon, ts, source, msg, value, unit, radius
- The twin polls every 20 s and re-runs the full decision analysis on every new alert
- No other code changes needed — the integration hook is already built

11. HONEST LIMITATIONS

The platform is designed to be transparent. The following limitations are known and clearly labelled in the interface:

■ Network geometry is schematic

Pipe positions are from Civil 3D plan-view drawings — horizontal alignment is accurate but vertical depth uses the invert/rim levels, not true survey coordinates. The 3D height exaggeration is intentional for visibility.

■ Trunk main is schematic (~23 m grid)

The trunk main is modelled from the DXF plan — it is accurate in route but may not reflect as-built final positions.

■ Population is planning-density based

Population in the buffer is computed as plot area times an HMDA planning density per zone type. It is not census data and will overestimate in not-yet-built areas.

■ Burst discharge assumes 30 m head

The orifice equation uses an assumed 30 m static head. The true discharge depends on the actual operating pressure, which requires a live pressure sensor.

■ Sensor readings are virtual

All ~50 sensors are simulated. They follow real diurnal patterns and fault physics but produce no real operational information. Labelled ■ SIM throughout.

■ Flood depth not modelled

A DEM is required for true flood depth simulation. The current storm module assesses drain capacity and surcharge risk but cannot predict water depth.

■ Building position is approximate

The WTP Admin Building is placed at the network centre as a georeferenced placeholder. The exact position requires the surveyed Easting/Northing from the site plan.

■ ECW ortho not integrated

The Zone 5D ortho image is in ECW (proprietary Hexagon format) which cannot be decoded without the Hexagon SDK. Conversion to GeoTIFF in QGIS is required first.

■ Internet required for map + weather

The 3D viewer and core analysis run offline. The Esri satellite basemap, Open-Meteo weather and Windy radar require an internet connection.

■ Decision engine is not trained AI

The recommended actions come from deterministic engineering rules, not a machine-learning model. Confidence levels are rule-based, not statistical. ML is possible after approximately one year of accumulated SCADA sensor history.

12. CREDITS & CONTACT

Client	Andhra Pradesh Capital Region Development Authority (APCRDA) Amaravati, Andhra Pradesh, India
Platform developer	Kreayotoo Solutions (P) Ltd (KSPL) Hyderabad, Telangana — Engineering · Solutions · Innovation
Data sources	Civil 3D utility network quantities (APCRDA/contractor workbooks) Revit IFC4 architectural + structural models (WTP Admin Building) APCRDA Masterplan shapefile (95,935 plots, landuse zoning) Open-Meteo (weather API) · Windy.com (radar embed) Esri World Imagery (satellite basemap)
Standards & references	IFC4 (ISO 16739) · EPSG:32644 (UTM Zone 44N) · EPSG:4326 (WGS84) Manning's equation (n=0.014) · Rational Method (Q=CiA) CPHEEO Manual on Water Supply · HMDA Planning Norms WHO Guidelines for Drinking-water Quality (chlorine 0.2–0.5 mg/L) Siemens SIWA (operational benchmark reference)
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This report documents the Amaravati Capital City Digital Twin built by Kreayotoo Solutions (P) Ltd. All data from APCRDA sources is used solely for this project. The simulated sensor data is clearly labelled and does not represent actual operational readings. This document is confidential.